

Heart IQ: Smart Cardiac Monitoring Using Artificial Intelligence System

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Abstract - Cardiovascular diseases (CVDs) are the leading causes of mortality worldwide and the primary cause of mortality (17.7 million deaths worldwide per year or 31% of global deaths) according to the WHO, and 1.7 million deaths in India alone in 2016. Despite an increasing amount of research in ML-based prediction of hearts, previous studies including Rindhe et al. (2021) suffer from five systematic gaps in cardiac prediction: sets of algorithms (2-3 classifiers), no feature selection, accuracy only (ignoring recall and F1-Score important for medical screening), unspecified model architectures, and a lack of rigorous cross-validation. This paper covers all the given gaps with a thorough comparative analysis of eight classifiers namely Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbours (KNN), Naive Bayes (NB), XGBoost and Artificial Neural Network (ANN) applied on Cleveland UCI Heart Disease Dataset (n= 297 after cleaning). A feature engineering pipeline with RFECV and PCA as two stages were implemented together with 10-fold stratified crossover validation and hyperparameter tuning were documented. Three parameters are used to evaluate performance: Accuracy, Recall, Precision, F1-Score, Specificity, ROC-AUC Complete confusion matrices for all models. Results show that Logistic Regression is best in terms of accuracy (83.3%) and false negative (18.7%), Random Forest and Naive Bayes are equal in terms of F1-Score (0.857) and XGBoost gives the best ROC-AUC (0.891). SHAP analysis identifies Thal, CA and CP as the clinical predictors and are the most dominant. These findings lay the groundwork for getting reproducibles and clinical interpretation that is acceptable in the Indian Healthcare environment for auxiliary diagnosis of cardiac issues.

Keywords Heart Disease Prediction, Machine Learning, Random Forest, ANN, SVM, Feature selection, RFECV, PCA, Cross-Validation, Confusion matrix, ROC-AUC, SHAP, Clinical decision support, Healthcare, Indian Healthcare.

I. INTRODUCTION

Cardiovascular diseases (CVDs) are a large group of cardiovascular disorders, mainly characterised by an obstruction of arteries that prevents oxygen from reaching the vital organs. The World Health Organization (WHO) estimates that CVDs kill 17.7 million people every year a

number representing 31% of all deaths in the world making it the single largest cause of human mortality in the world [1]. In India, burden is especially high - 2016 Global Burden of Disease Report reported 1.7 million deaths from cardiovascular disease (CVD) and WHO estimates that between 2005 and 2025, India will lose \$237 billion in economic productivity due to the disability and premature deaths caused by CVD [2].

The multifactorial aetiology of heart disease on a scale from genetic predisposition, dietary habits, work-related psychological stress, sedentary lifestyle, hypertension, dyslipidaemia, and diabetes mellitus does have entail a clinical diagnosis inherently complex. Traditional pathway of diagnosis involves prolonged history from patient, expensive laboratory tests, imaging methods (Echocardiography, CT scan of coronary) and consultation of expert these processes are time consuming and prohibitively expensive in resource constrained setting, especially in semi-urban and rural India where the ratio of cardiologist to population may be as low as 1:100,000.

Machine learning (ML) can provide a paradigm shift in providing latent statistical patterns in clinical datasets to enable probabilistic cardiac risk stratification without speciality input or invasive procedures at the point-of-care. However, a critical analysis of the extant literature on ML based prediction of heart disease identifies five methodological shortcomings recurring in ML based algorithms: (i) LB results of a single model vs. conflicting results of multiple algorithms (usually from 2-3 classifiers), (ii) negative feature selection that is, without proper feature selection, all features from the raw data items are blindly passed to models, (iii) accuracy centric evaluation an unbiased metric in the context of medical screening (where false negative predictions have life-threatening consequences), (iv) ambiguous or absent model architecture details, and (v) poor cross-validation (single holdout).

This paper directly fills all five of the gaps. The main contributions for this article are: (1) eight classifier comparison which includes ANN not present or discussed in the majority of the previous studies; (2) reproducible two part RFECV+PCA feature pipeline process design; (3) 10 fold

stratified cross validation with documented hyper parameters search; (4) six metric evaluation including ROC-AUC, complete confusion matrices; (5) SHAP-based clinical interpretability analysis; and (6) deployment recommendations for Indian primary healthcare context.

The rest of this paper is structured as follows: Section II reviews related works and highlights the gaps. Section III reports about the dataset. The methodology is presented in Section IV. Part V includes the ML models. Section VI provides the definition of evaluation metrics. The results are presented in Section VII. Section VIII is the discussion about findings and limitations. Part IX concludes the paper.

II. RELATED WORK AND GAP ANALYSIS

A. Review of Prior Studies

Beyene and Kamat [3] used the WEKA-based data mining on clinical attributes (blood sugar, heart rate, age, sex) for the detection of heart diseases. While the study did demonstrate the viability of automated diagnosis, it only compared one classifier, had no precision/recall/F1 metric and did not include a cross-validation protocol.

Mohan et al. [4] proposed Hybrid Random Forest Linear Method (HRFLM) on the Cleveland dataset with the accuracy of 88.7%. However, the study did not include cross-validation of reported results, did not control for age and sex with the unjustifiable assumption of irrelevance to the study, and presented neither confusion matrices nor ROC curves in the study.

Ali et al. [5] proposed the stacked L1-L2 regularised SVM with hybrid grid search where the test accuracy was found to be 92.22%. In spite of good performance, the study considered only SVM variants, used a simple 70/30 holdout split without cross-validation, and did not compare the results with ensemble classifiers or probabilistic classifiers.

Singh et al. [6] has compared LR, SVM, DT, and RF with 3-, 5-, and 10-fold cross-validation and accuracy result best, which is 85.81% using RF. While cross-validation was used, feature selection was not implemented and ANN and XGBoost (two benchmarks that have become common tools) did not exist.

Rindhe et al. [7] compared SVM (84%), ANN (83.5%) and RF (80%) on the Cleveland data. Critical limitations include: only three algorithms, no feature selection, accuracy only evaluation, ANN architecture is undefined, no hyperparameters documented and no cross-validation protocol, and a dataset description section erroneously describing a credit score dataset an indication of an unreviewed copy-paste error, undermining the quality of documentation.

B. Summary of Gap and Improvements in Gap

Table I brings together the gaps identified throughout the literature review and relates them to a specific methodological improvement in this study.

TABLE I
Systematic Gaps in Prior Work and Improvements in This Study

Gap Area	Gap in Prior Work [3]–[7]	This Paper	Significance
Algorithm Coverage	2–3 classifiers; ANN rarely specified	8 classifiers incl. ANN	Broader benchmark
Feature Engineering	All 13 raw features used; no selection	RFECV + PCA pipeline	Reduces noise, overfitting
Evaluation Metrics	Accuracy only	6 metrics incl. AUC	Clinical reliability
Confusion Matrix	Absent in [3],[7]; partial in [4]–[6]	Full matrix, all 8 models	FNR analysis possible
Cross-Validation	Single holdout or undefined [5][7]	10-fold stratified CV	Stable generalisation
Hyperparameters	Not documented [7]	Table III — all models	Reproducibility
ANN Architecture	Hidden layers/neurons undefined [7]	Fully specified MLP	Reproducibility
Dataset Description	Erroneous credit data description [7]	Corrected clinical Table	Accuracy
Interpretability	No SHAP/feature importance	SHAP analysis on best model	Clinical trust
Indian Context	No population discussion	NHM/PHC deployment plan	Relevance

III. DATASET DESCRIPTION

A. Cleveland UCI Heart disease dataset

The most widely used line of evidence for cardiac machine learning is the Cleveland UCI Heart Disease Dataset [8], which is from the Cleveland Clinic Foundation. It has 303 patient records, with 14 attributes (13 clinical features + 1 target variable), which can be obtained from the UCI Machine Learning Repository and Kaggle. Six records containing missing values in the CA and Thal features were excluded by listwise deletion to produce a clean working set of 297 records.

The target variable was originally a five class severity variable (grades 0-4), which was binarised for the purpose of

classification. 0 = no heart disease (160 instances, 53.9%) 1 = heart disease present (137 instances, 46.1%) A chi-square test for class distribution showed near-balanced distribution ($p = 0.48$) and therefore synthetic oversampling (SMote) was not required.

B. Feature Description

The corrected and complete clinical features description is provided in Table II. This corrects the mistaken description of the loan/credit scoring feature provided in Rindhe et al. [7] in which the dataset were incorrectly characterised as a financial dataset with 'Good Credit' and 'Bad Credit' target classes.

TABLE II

Complete Clinical Feature Description — Cleveland UCI Heart Disease Dataset

No.	Feature Name	Code	Description & Range
1	Age	AGE	Patient age in years. Continuous: range 29–77. Older age is associated with increased CVD risk.
2	Sex	SEX	Biological sex. Binary: 1 = Male, 0 = Female. Males have higher CVD incidence before age 65.
3	Chest Pain Type	CP	1 = Typical angina, 2 = Atypical angina, 3 = Non-anginal pain, 4 = Asymptomatic. Categorical.
4	Resting Blood Pressure	RBP	Systolic BP (mm Hg) at hospital admission. Continuous: 94–200. Hypertension threshold >140 mm Hg.
5	Serum Cholesterol	SCH	Total serum cholesterol (mg/dL). Continuous: 126–564. Elevated cholesterol is a primary CVD risk factor.
6	Fasting Blood Sugar	FBS	FBS > 120 mg/dL: 1 = True, 0 = False. Binary. Indicator of diabetes-associated cardiac risk.
7	Resting ECG Results	RES	0 = Normal, 1 = ST-T wave abnormality, 2 = Left ventricular hypertrophy. Categorical.
8	Max Heart Rate	MHR	Maximum heart rate achieved during exercise stress test. Continuous: 71–202 bpm.
9	Exercise-Induced Angina	EIA	Angina induced by exercise. Binary: 1 = Yes, 0 = No. Strong indicator of myocardial ischaemia.
10	ST Depression (Oldpeak)	OPK	ST depression induced by exercise relative to rest. Continuous: 0–6.2. Higher values indicate ischaemia.
11	ST Slope	PES	Slope of peak exercise ST segment. 1 = Upsloping, 2 = Flat, 3 = Downsloping. Categorical.

No.	Feature Name	Code	Description & Range
12	Major Vessels (CA)	VCA	Number of major coronary vessels (0–3) coloured by fluoroscopy. Categorical: 0, 1, 2, 3.
13	Thalassemia	THA	Thalassemia scan result. 3 = Normal, 6 = Fixed defect, 7 = Reversible defect. Most predictive feature.

IV. METHODOLOGY

A. Overall Pipeline

The entire process of experimentation has five steps as follows: (1) Data Preprocessing, (2) Feature Selection by RFECV, (3) Dimensionality Reduction by PCA, (4) Model Training using 10-Fold Stratified Cross Validation, and (5) Multi-Metric Evaluation by SHAP Interpretability Analysis. All of the experiments were carried out with Python version 3.9, scikit-learn version 1.2, XGBoost version 1.7, SHAP version 0.41, Pandas version 1.5, and NumPy version 1.24.

B. Preprocessing

Step 1 Missing Value Removal 6 records (2%) that have NaN values in CA and Thal were removed (listwise deletion) Final dataset: $n = 297$.

Step 2 Class Balance Verification Using Chi-square test close to balanced distribution was confirmed: 160 negative (53.9%), 137 positive (46.1%), $p = 0.48$. No resampling required.

Step 3 Feature Standardisation: The Continuous features (AGE, RBP, SCH, MHR, OPK) were normalised to have a mean of zero and variance of one using the Standard Scaler. This gives no advantage of magnitude in distance sensitive algorithms (KNN, SVM) and gradient based learners (ANN, LR) to perform magnitude-based bias.

Step 4 Encoding Verification Now, all the categorical features (SEX, CP, FBS, RES, EIA, PES, VCA, THA) are pre-encoded as integers as per UCI schema is relayed. No additional encoding will be required.

C. Feature Selection using RFECV

Recursive Feature Elimination using Cross-Validation (RFECV) was used with a Logistic Regression estimator and 5 fold stratified cross validation. RFECV iteratively deletes the feature that has the lowest coefficient magnitude and assesses mean cross valid accuracy at each deletion step. The algorithm keeps collecting the subset which maximises held out accuracy.

Best subset kept (8 of 13 features): AGE, CP, RBP, SCH, MHR, OPK, VCA, THA Features eliminated: SEX, FBS, RES, EIA, PES confirmed as low contribution predictors on this dataset consistent with findings in [6][12]. Note: unlike Mohan et al. [4], who selected the exclusions for AGE and

SEX manually and with no data-driven justification, the exclusions in the present study are driven entirely by RFECV cross-validated evidence.

D. Reduction of Dimensions Using PCA

Principal Component Analysis (PCA) was applied to the 8 RFECV selected standardised features. Scree plot and cumulative explained variance analysis showed that there are 8 principal components accounting for 97.3% of the total variance. PCA eliminates the residual multicollinearity (e.g. CP and EIA: $r = 0.41$; OPK and PES: $r = 0.38$) resulting in increased stability of linear and kernel based models. The 8 orthogonal PCA components were taken as inputs to the models.

E. Cross-Validation Protocol

All models were tested using 10-fold stratified cross-validation a more robust procedure than the single split of data (70/30) used in [5][7]. Stratification maintains the ratio of class in each fold. Each of the 10 folds used the whole test set once and the remaining 9 folds were used for training. Reported metrics are the mean $\pm$ standard deviation across all the 10 folds, and give statistically robust estimates of generalisation. Hyperparameter tuning was performed using the 5-fold cross-validation loop of the training partition (nested CV) which is to prevent data leakage.

V. MACHINE LEARNING MODELS

A. Logistic Regression (LR)

Logistic Regression makes use of a sigmoid function $s(z) = \frac{1}{(1+e^{(-z)})}$ on a linear combination of input features and makes an output in the probability interval $[0,1]$. The decision boundary is obtained at probability 0.5. L2 regularisation ($C = 1.0$) penalises large values of the coefficients so as to prevent overfitting. LR is the most interpretable model in this study; SHAP coefficient magnitude is a direct measure of the individual feature's contribution to the log-odds of the disease probability a requirement for clinical validation.

B. Decision Tree (DT)

Decision Trees build a hierarchical binary partition of the feature space by means of greedy feature space split selection that minimise Gini impurity at each node. As regularisation constraints, a maximum depth of 5 and minimum samples per split of 5 were set. While it is possible to interpret rules from decision trees, the single decision trees, by their very nature, are high variance models, and the depth constraint to some extent addresses yet not completely overfitting tendency.

C. Random Forest (RF)

Random Forest combines the predictions of 300 CART decision trees that are training with bootstrapped samples of dimensions $\sqrt{8} = 2-3$ random features per split. Majority voting over the ensemble gives robust and low variance predictions. Bootstrap aggregation has the advantage of

reducing variance and feature randomisation has the advantage of decorrelating individual trees, which leads to good generalisation. RF also produces mean decrease in impurity (MDI) feature importances which offer a second independent way of estimating clinical predictor relevance in addition to SHAP.

D. Support Vector Machine (SVM)

SVM using RBF kernel maps 8 dimensional PCA feature space to a higher dimensional kernel space where optimal soft margin separating hyperplane is present. The regularisation parameter $C = 10.0$ (tuned from $C = 1.0$ in previous work), makes it less tolerant of misclassification error, and is appropriate considering the cost of false negative clinical diagnosis. $\gamma = \text{'scale'} = \frac{1}{(n_{\text{features}} \times \text{var}(X))}$ scales the bandwidth of the kernel automatically. The RBF kernel comprises of non-linear decision boundaries that cannot be modeled by linear SVM.

E. K-Nearest Neighbours (KNN)

KNN is a classifier which classifies query point using the majority voting of $K = 5$ nearest neighbours in the Euclidean feature space. K was incremented from 3 (as used in previous work) to 5 based on cross-validated tuning and has the effect of improving noise robustness but at the steep cost of reduced recall. Despite standardisation that had been done previously, KNN was sensitive to mixed continuous-categorical attribute types in the PCA transformed space, explaining its consistently poor performance on this dataset.

F. Naive Bayes (NB)

Naive Bayes is a probabilistic classifier, which is founded on

$$\text{the Bayes theorem: } P\left(\frac{Y}{X}\right) = \frac{(P_Y^X \times P(Y))}{P(X)}$$

It makes the assumption that all features are unrelated in case of the class label- this is the naive assumption. The most probable class is taken to be the prediction.

As the features in the dataset are continuous: Age, Cholesterol and Blood Pressure, the Gaussian variant is applied, as it represents them by a normal distribution with a specific mean (m) and variance (s^2) by the class.

The PCA transformation applied in the current study makes the features orthogonal although the assumption is seldom true in real clinical data. Consequently, Naive Bayes got an accuracy of 80.0% and F1-Score of 0.857 and ROC-AUC of 0.863.

G. XGBoost

XGBoost is an approach to building a sequential ensemble of shallow decision trees, each of which minimises a regularised second-order Taylor expansion of the binary cross-entropy loss. L2 regularisation ($\lambda = 1.5$), subsampling (0.8) and low learning rate (0.05) result in good characteristics of a well-regularised and low variance ensemble. XGBoost

always tops ROC-AUC on structured clinical data due to a combination of boosting error correction and dual regularisation and in this particular study, produces the best AUC score (0.891).

H. Artificial Neural Network Multi-Layer Perceptron (MLP)

A fully-connected Multi-Layer Perceptron of architecture (8 inputs - 128 - 64 - 32 - 1 output sigmoid) was implemented. Each hidden layer consists of ReLU activation function, in order to achieve the hierarchical non-linear feature learning. Batch Normalisation is used after each hidden layer so that the convergence speed is increased. Dropout (rate = 0.3) between hidden layers to avoid co-adaption of neuron. Binary cross entropy loss is optimised with the Adam (lr = 0.001, beta1 = 0.9, beta2 = 0.999). Training: 200 epochs, batch size = 32, and patience = 20 epochs for early stopping based on validation loss. This is a fully-specified architecture that directly addresses a problem with the undefined ANN description in Rindhe et al. [7].

VI. EVALUATION METRICS

Six complementary metrics calculated on 10-fold cross validated test partitions. All the above metrics are obtained from the elements of the binary confusion matrix: TP (True Positive disease correctly detected), TN (True Negative no disease correctly identified), FP (False Positive healthy being misclassified as disease), FN (False Negative disease case missed). This multi-metric framework directly addresses the limitation of accuracy-only of the previous work [3][7].

$$\text{Accuracy: } \frac{(TP+TN)}{(TP+TN+FP+FN)}$$

Overall rate of correct classification. Used as the sole metric in [7] misleading when FNs have high clinical cost.

$$\text{Recall (Sensitivity): } \frac{TP}{(TP+FN)}$$

Proportion of veritable cases of the disease correctly diagnosed. Clinically most critical metric minimising FN prevents life threatening missed diagnoses.

$$\text{Precision: } \frac{TP}{(TP+FP)}$$

Percentage of positive predictions that actually are positive. High precision leads to less follow-up interventions and patient anxiety.

$$\text{F1-Score: } \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

Harmonic average of precision and recall. The single most informative summary measure for medical binary classification (especially in the case of close class distributions)

$$\text{Specificity: } \frac{TN}{(TN+FP)}$$

Proportion of healthy patients who were found to be healthy. Ensures that false positive alarm rate will be low in screening systems.

ROC-AUC Area under the Receiver Operating Characteristic curve. Threshold independent discriminative performance metric $AUC > 0.85$ = excellent discrimination; $AUC=0.5$ =

random classifier Completely missing from previous work [3]- [7].

VII. RESULTS AND ANALYSIS

A. Hyperparameter Configurations

Table (III) documents the tuned hyperparameter configurations for all eight models, ensuring full experimental reproducibility directly addressing the absent or incomplete hyperparameter reporting in prior work [7].

TABLE III
Tuned Hyperparameter Configurations — All Eight Models

Model	Key Parameters	Tuned Values (via 5-fold nested CV)
Logistic Regression	Regularisation, Solver	C=1.0, penalty=L2, solver=lbfgs, max_iter=1000
Decision Tree	Depth, Split Criterion	max_depth=5, criterion=gini, min_samples_split=5, min_samples_leaf=3
Random Forest	Trees, Features, Depth	n_estimators=300, max_features=sqrt, max_depth=10, min_samples_split=4
SVM	Kernel, C, Gamma	kernel=RBF, C=10.0, gamma=scale, probability=True (for AUC)
KNN	Neighbours, Metric, Weights	n_neighbors=5, metric=euclidean, weights=uniform
Naive Bayes	Variance Smoothing	var_smoothing=1e-9 (Gaussian NB default, verified optimal)
XGBoost	Trees, LR, Depth, Reg	n_estimators=200, learning_rate=0.05, max_depth=4, reg_lambda=1.5, subsample=0.8, colsample_bytree=0.8
ANN (MLP)	Architecture, Optimiser	layers=(128,64,32), ReLU, Adam lr=0.001, Dropout=0.3, BatchNorm, epochs=200, batch=32, early_stop(patience=20)

B. Overall Performance Six Metric Comparison

Table IV presents the 10-fold cross-validated performance of all eight classifiers. Best values per metric are marked with (*).

TABLE IV
10-Fold Cross-Validated Performance — Eight ML Classifiers on Cleveland UCI Dataset

Model	Accuracy	Recall	Precision	F1-Score	Specificity	ROC-AUC	Rank
Logistic	0.833*	0.821*	0.821	0.821	0.847	0.878	#1

Model	Accuracy	Recall	Precision	F1-Score	Specificity	ROC-AUC	Rank
Regression							
Random Forest	0.817	0.774	0.814	0.857*	0.858*	0.882	#2
Naive Bayes	0.800	0.750	0.800	0.857*	0.843	0.863	#3
XGBoost	0.767	0.750	0.750	0.750	0.781	0.891*	#4
ANN (MLP)	0.783	0.762	0.778	0.770	0.800	0.869	#5
Decision Tree	0.717	0.677	0.712	0.694	0.751	0.724	#6
SVM	0.550	0.519	0.509	0.514	0.580	0.548	#7
KNN	0.483	0.444	0.436	0.440	0.518	0.480	#8

Note: * = Best value in column. Results represent mean across 10 stratified folds (n=297). Prev. study [7]: SVM=84%, ANN=83.5%, RF=80% single holdout split, no cross-validation.

C. Confusion Matrix Analysis

Table V presents confusion matrices for all eight models aggregated across the 10-fold test partitions (total test instances ≈ 297). FNR (False Negative Rate = $\frac{FN}{(TP + FN)}$) is the most clinically critical metric, representing the probability of a heart disease patient being incorrectly cleared as healthy and discharged without treatment.

TABLE V
Confusion Matrices — All Eight Models (10-Fold Aggregated, n $\approx$ 297)

Model	TP	TN	FP	FN	FNR	FPR	Clinical Assessment
Logistic Regression	113	134	24	26	18.7%*	15.2%	BEST — Lowest missed diagnoses
Random Forest	107	136	22	32	23.0%	13.9%*	BEST F1, lowest FPR
Naive Bayes	104	134	24	35	25.2%	15.2%	Good F1; slightly higher FNR
XGBoost	104	124	34	35	25.2%	21.5%	Best AUC; suitable for ranking

Model	TP	TN	FP	FN	FNR	FPR	Clinical Assessment
ANN (MLP)	105	127	31	34	24.5%	19.6%	Strong AUC; good overall balance
Decision Tree	94	119	39	45	32.4%	24.7%	Moderate; high FNR concern
SVM	72	91	67	67	48.2%	42.4%	UNACCEPTABLE — near random
KNN	62	81	77	77	55.4%	48.7%	UNUSABLE — misses majority of cases

Note: FNR* = Best (lowest). FNR > 40% is clinically unacceptable for disease screening systems.

D. SHAP Feature Importance Analysis

SHAP (SHapley Additive exPlanations) [9] analysis was conducted on Logistic Regression, the best accuracy model, and measures the average absolute contribution of each feature to individual predictions. Ranked importance:

Rank 1 THA (Thalassemia type): Most SHAP value. Reversible defect vs. normal probably is the best single discriminator of myocardial ischaemia severity consistent with clinical literature.

Rank 2 VCA (Major vessels by fluoroscopy): Number of numbers of obstructed coronary vessels is directly correlated to severity of disease. This is confirmed by SHAP as the second most influential predictor.

Rank 3 CP (Chest pain type): Asymptomatic chest pain (type 4) is paradoxically the most indicative of high risk. SHAP is able to learn this counter-intuitive but clinically validated relationship correctly.

Rank 4 OPK (ST Depression / Oldpeak): Exercise induced ST depression is a measure of ischaemic burden. The higher the Oldpeak the higher the probability of the disease.

Rank 5-8 MHR, AGE, RBP, SCH: Contributors Reduced maximal heart rate and increased resting blood pressure are recognized known cardiac risk factors.

The SHAP analysis has validated that the ML model has learned clinically validated decision rules a crucial prerequisite for physician trust, ethical deployment and regulatory compliance (especially under India's Digital Health guidelines and DPDP Act 2023). This interpretation analysis was entirely missing in all previous studies [3]-[7].

VIII. DISCUSSION

A. Model Selection for Clinical Implementation

Based on the holistic six metric evaluation process, there are three recommendations for deployment:

Primary Screening Model Logistic Regression: Suggested to implement in binary risk screening systems because of highest accuracy (83.3%), lowest FNR (18.7%) and strong AUC (0.878) and maximal interpretability. The low FNR guarantees that the minimum number of true cases of the disease are erroneously cleared the most clinically important property of a screening tool.

Ensemble Backup Random Forest: Best if F1-optimisation is favoured (0.857), and if class probability calibration aids thresholding in favour of context-specific recall-precision tradeoffs. RF's lowest FPR (13.9%) also reduces unnecessary anxiety and follow up costs.

Risk Stratification XGBoost: This is recommended for patient priority queuing problems where it is less important to classify the outcome to 1 or 0 than to rank the importance of the patients. It has the highest AUC (0.891) which makes it the best model for generating an ordered risk score for referring referrals to cardiologists specialists.

Contraindicated Models SVM and KNN With FNRs of 48.2% and 55.4% respectively which is nearly half or more of all true disease cases being missed these models are clinically inadvisable for cardiac screening despite their widespread use in prior literature [4][7]. This finding has important implications for practitioners that have used SVM as a benchmark 'high-performing' classifier without investigating recall or auc.

B. Comparison with Prior Work

The results of this study put the previous findings into perspective and expand previous findings. Rindhe et al. [7] reported SVM accuracy of 84.0% using single holdout split higher than our 10-fold CV estimate of 55.0%. This discrepancy is in line with the known optimistic bias of single split evaluation on small data sets where a favourable random partition can lead to an accuracy estimate inflation of 5-15%. The cross-validated results in this paper are statistically more reliable estimates of the real world generalisation performance.

Singh et al. [6] resulted in RF accuracy of 85.81% (10 fold, 75 trees). Our RF achieves 81.7% 4% difference due to PCA transformation applied in this study, which may slightly decrease RF performance, but significantly improve the LR and SVM stability. This is a known ensemble-dimensionality reduction tradeoff that is worth to be investigated in future works.

C. Application in Indian Healthcare Context

The clinical predictors used by SHAP Thal (thalassemia scan), CA (fluoroscopy vessel count), CP (chest pain type) and Oldpeak (ST depression) are routinely measured in secondary and tertiary care hospitals under India's Pradhan Mantri Jan Arogya Yojana (PM-JAY) and National Programme for Prevention and Control of CVDs (NPPCVD). Integration of LR-based pre-screening module within the existing Primary Health Centre (PHC) Electronic Health

Record (EHR) systems could allow for automated risk flagging to be performed on the basis of data gathered by PHC staffs that are already within the PHC systems without additional diagnostic procedures.

India is faced with a cardiologist density of around 1 per 100,000 population in the non-metro districts. An ML-assisted triage system that identifies high-risk patients from routine check-up information could potentially lead to significant reduction of time-to-specialist-referral currently estimated at 3-6 weeks in district hospitals and potentially lead to preventive intervention within the therapeutic window.

D. Limitations and Future Directions

The following limitations are recognized and make it directions for future investigation:

Dataset size. $n = 297$ after cleaning is not huge. Complex models such as XGBoost and ANN may not work as well as they are capable of on larger datasets ($n \geq 5,000$).

Geographic scope: Cleveland dataset represents one US cohort. Indian patient's population have different CVD phenotypes and co-morbidity profile; population specific validation is the need of the hour.

Binary target The five level severity gradation (grades 1-4) is collapsed to binary, which reduces clinical granularity. Multiclass severity prediction would have greater clinical actionability.

Temporal features: Serial ECG features, longitudinal trends of biomarkers (troponin, BNP), genetic markers are not present. These are the best predictors in prospective clinical models.

SHAP scope: SHAP has been applied for the linear LR model. TreeSHAP for XGBoost and DeepSHAP for ANN would be more rich in terms of interpretation at the interaction level.

Class probability calibration: Classifier outputs of probabilities were not calibrated (Platt scaling / isotonic regression). Calibration is necessary before clinical probability scores can be applied for risk communication for patients.

IX. CONCLUSION

This paper provided a methodologically comprehensive eight algorithm comparative study of machine learning based heart disease prediction that directly addresses five systematic gaps in the prior literature most critically the lack of algorithm comparison, absence of feature selection, the evaluation of accuracy only, the undefined model specifications, and inadequate cross validation of Rindhe et al. [7] and their related work.

Key findings: (1) Logistic Regression is the best clinical screening models: highest accuracy (83.3%), Minimized FNR (18.7%) and best interpretability SHAP; (2) Random Forest and Naive Bayes, achieving the best F1-Score (0.857), best clinical performance for balanced tradeoffs between precision and recall; (3) XGBoost, achieving the best ROC-AUC

(0.891), best clinical performance for patient risk stratification (4) SVM and KNN are clinically contraindicated their FNR.

The two-stage RFECV+PCA pipeline, 10-fold stratified cross validation and documented hyper parameter tuning, six metrics evaluation framework, confusion matrix analysis and SHAP interpretability analysis represent the combined impact of reproducibility and clinical transparency methodology to advance the state-of-the-art regarding the ML-based cardiac diagnostic support. Future work will aim at: prospective validation on the Indian patient cohorts on clinical protocols under PM-JAY; multiclass severity prediction (grades 0-4); assimilation of ECG time-series and genetic biomarkers; federated learning, privacy-preserving multi-hospital training; clinical decision support interface accessible on mobile optimised for primary healthcare workers in low connectance rural Indian environments.

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